

Gregory Bian

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Computer Engineering (Co-op)

Relevant Courses: Circuit Analysis, Computing Systems, Software Construction, Data Structures & Algorithms

Expected May 2029

Vancouver, BC

TECHNICAL SKILLS

Firmware & Embedded Systems: CAN, SPI, I2C, UART, FreeRTOS, STM32, FPGA

Programming & Frameworks: C, C++, Python, Java, JavaScript (React), SystemVerilog

Systems & Tools: Git, Docker, Linux (Ubuntu), Pytest, JUnit

ENGINEERING STUDENT DESIGN TEAMS

Firmware Developer

Sep 2025 – Present

UBC Solar Racing

Vancouver, BC

- Collaborating with a cross-disciplinary team of 80+ software, electrical, and mechanical engineers, writing firmware in C for a solar-powered car, Cascadia, in preparation for FSGP and ASC solar racing competitions.
- Coordinated with the BMS and Power & Signals subteams to calibrate the accelerator pedal across a CAN Bus to the motor, implementing fail-safes, and **ensuring precise and reliable control in a safety-critical system.**
- Led the design and implementation of a multi-page LCD Screen over a STM32 microcontroller, consulting previous drivers for layout feedback and **improving visibility of real-time driving and diagnostic data by 30%.**
- Implemented memory-robust, FreeRTOS-based task scheduling on the LCD screen to **reduce latency by 20%** and enable concurrent operation of LCD updates with driver lights and turn signal updates.

Software Developer

Jan 2025 – Mar 2025

UVIC Aero

Victoria, BC

- Configured a Linux-based flight simulator to test autonomous flight functions for Firefly, a fire-extinguishing drone; documented WSL and terminal commands to **enable knowledge transfer to future team members.**
- Engineered a Python-based hotspot detection algorithm for infrared images using thresholding and connected components analysis in OpenCV, identifying over 85% of high-intensity regions in test datasets of 30+ images.

WORK EXPERIENCE

STEM and Computing/Robotics Instructor

Jun 2025 – Aug 2025

UBC Geering Up Outreach

Vancouver, BC

- Led STEM-based Summer Camps for Geering Up, a STEM Outreach Program at the Faculty of Applied Science at UBC, **reaching over 28000+ youth a year.**
- Instructed up to 30 students per week on Arduino-based embedded programming, computer architecture, and data analysis in Python, explaining complex technical concepts in a fun and engaging way for youth.

PROJECTS

NotiFlow Course Tracker | *JavaScript (React), Flask, Python*

Nov 2025 – Dec 2025

- Led a team of 5 in developing a Course Tracking Chrome Extension; implemented a React-based UI and integrated Chrome Local Storage APIs to store data locally, improving usability and reliability.
- Leveraged Flask routing to send and receive data between the Chrome Extension and backend Python scripts and support features such as Canvas API access, BeautifulSoup webscraping and OpenAI API prompt filtering.

Bare-Metal Tron Game | *C, RISC-V Assembly, FPGA*

Nov 2025

- Developed a bare-metal Tron game in C, generating VGA output for real-time graphics rendering on a DE10-Lite FPGA and featuring multiplayer or single-player against a simple logic-driven robot.
- Implemented hardware interrupt-driven control logic using RISC-V Assembly to handle player and robot movement, collision detection (deaths), and real-time HEX display updates.

Hand Hygiene Compliance System | *MicroPython, JavaScript, Raspberry Pi*

Mar 2025 – Apr 2025

- Engineered a prototype embedded system with Raspberry Pi in MicroPython to monitor Hospital Hand Hygiene Compliance of workers, resulting in a 30% higher efficiency compared to alternate designs.
- Developed a full-stack user interface in HTML/CSS/JavaScript, featuring real-time compliance data visualization and an interactive reset button with two-way communication over Wi-Fi to backend systems.